RANDOM VARIABLES

In a Random experiment if a real variable is associated with every outcome, then it is called a random variable or stochastic variable. In other words, a random variable is a function that assigns a real number to every sample point in the sample space. Random variables are usually denoted by capital letters. It may be noted that diff. random variables may be associated with same sample space. The set of real no.s. of a random variable is called the range of X.

Suppose a coin is tossed twice, we shall associate to diff ~~kind of~~ random variables X and Y as follows:

X = Number of heads in the outcome

Y = Number of tails in the outcome

We have $S = \{HH, HT, TH, TT\}$

The association of elements in S to X is as follows

Outcome	HH	HT	TH	TT
X	2	1	1	0

Illy the association of ele in S to Y

Outcome	HH	HT	TH	TT
Y	0	1	1	2

∴ The range of $X = \{0,1,2\}$ and of $Y = \{0,1,2\}$

$P(X=0) = \frac{1}{4}$ $P(X=1) = \frac{2}{4} = \frac{1}{2}$ $P(X=2) = \frac{1}{4}$

Discrete and Continuous random variables:

If a random variable takes finite or countable infinite number of values, then it is called discrete random variables.

If a random variable takes non-countable infinite no. of values, then it is called continuous random variables.

Eg:- Tossing the coin and observing outcome

Throwing a die and observing no. of on the face

• Observing a phr on a speedometer (continh. random
• conducting a survey on life of electric bulbs.

Discrete probability distribution

If for each value x_i for a discrete random variable X, we assign a real no. $p(x_i)$ such that

i) $p(x_i) \geq 0$

& ii) $\sum_i p(x_i) = 1$

Then the funct^n $p(x)$ is called a probability function.

If the prob. that X takes the values x_i is P_i then $P(X = x_i) = P_i$ or $P(x_i)$

The set of values $\{x_i, p(x_i)\}$ is called a discrete probability distribution. The function $p(X)$ is called the probability density function.

Eg: 1) From a sealed box containing a dozen apples, it was found that 3 apples were perished. Obtain the prob distribution of the no. of perished apples when 2 apples are drawn at random.

Soln:- Let X be the no. of perished apples.

Since there are 12 apples, 2 apples can be drawn in $^{12}C_2$ ways.

$\therefore X = \{0, 1, 2\}$

Probability of getting 0 perished apples is

$$P(X=0) = \frac{^3C_0 \times ^9C_2}{^{12}C_2} = \frac{1 \times \frac{9 \times 8^2}{2\times 1}}{\frac{12 \times 11}{2\times 1}} = \frac{36}{12 \times 11} = \frac{6}{11}$$

Prob. of getting 1 perished apple is

$$P(X=1) = \frac{^3C_1 \times ^9C_1}{^{12}C_2} = \frac{3 \times 9}{\frac{12 \times 11}{2}} = \frac{9}{22}$$

Prob. of getting 2 perished apples is

$$P(X=2) = \frac{^3C_2 \times ^9C_0}{^{12}C_2} = \frac{\frac{3\times 2}{2} \times 1}{\frac{12\times 11}{2}} = \frac{1}{22}$$

Probability distribution

x	0	1	2
$P(x)$	$\frac{6}{11}$	$\frac{9}{22}$	$\frac{1}{22}$

1) Find the value of 'k' such that the following distribution represents discrete probability distribution. Hence find its mean and std deviation. Also find
i) $P(x<=1)$ ii) $P(x>1)$ iii) $P(-1<x\leq 2)$

x	-3	-2	-1	0	1	2	3
$f(x)$	k	2k	3k	4k	3k	2k	k

Since $\{x, p(x)\}$ is a discrete (finite) probability distribution.

we must have $p(x_i) \geq 0$ which will be true only if $k \geq 0$.

& $\sum_{i=1}^{7} p(x_i) = 1$

$k + 2k + 3k + 4k + 3k + 2k + k = 1$

$16k = 1$

$\therefore k = 1/16$

Mean $(\mu) = \sum x_i p(x_i)$

$= -3k - 4k - 3k + 0 + 3k + 4k + 3k = 0$

variance $(\sigma^2) = \sum (x_i - \mu)^2 p(x_i)$ $\quad \mu = 0$

$= 9k + 8k + 3k + 0 + 3k + 8k + 9k$

$= 40k = 40 \times \frac{1}{16} = \frac{10}{4} = \frac{5}{2}$

$= 2.5$

Std deviation $(\sigma) = \sqrt{v} = \sqrt{2.5}$

i) $P(x \leq 1) = \sum_{i}^{5} p(x_i) = k + 2k + 3k + 4k + 3k = 13k = \frac{13}{16}$

ii) $p(x>1) = p(x=2) + p(x=3) = 2k + k = \frac{3}{16}$

iii) $p(-1 < x \leq 2) = p(x=0) + p(x=1) + p(x=2)$

$= 4k + 3k + 2k = \frac{9}{16}$

2) A random variable X has $p(x) = 2^{-x}$ $x = 1, 2, 3, 4, \ldots$ S.T. $p(x)$ is a probability function. Also find i) $p(x$ is even)

ii) $p(x$ being divisible by 3) iii) $P(X \geq 5)$

1/4 1/16

Soln:- Clearly $p(x) = 2^{-x} = \frac{1}{2^x} \geq 0$ for $x = 1, 2, 3, 4, 5, \ldots$

geometric series.

$r = \frac{1}{2}$, $a = \frac{1}{2}$

$$\sum_{x=1}^{\infty} p(x) = \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \cdots$$

$$= \frac{a}{1-r} = \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1$$

$\therefore p(x) = 2^{-x}$ is a probability function.

i) $P(x \text{ is even}) = P(x=2) + P(x=4) + P(x=6) + \cdots$

$$= \sum_{x=2,4,6} p(x_i) = \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \cdots$$

This is a geometric series with common ratio, $r = \frac{1}{2^2} = \frac{1}{4} < 1$

$\therefore P(x \text{ is even}) = \frac{a}{1-r} = \frac{\frac{1}{4}}{1-\frac{1}{4}} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$

ii) $P(X \text{ being divisible by } 3) = P(x = 3, 6, 9, 12, \ldots)$

$$= \sum_{x=3,6,9} p(x_i) = \frac{1}{2^3} + \frac{1}{2^6} + \frac{1}{2^9} + \cdots$$

This is a geometric series with $r = \frac{1}{2^3} = \frac{1}{8} < 1$

$\therefore P(X \text{ divisible by } 3) = \frac{\frac{1}{8}}{1-\frac{1}{8}} = \frac{1}{7}$

iii) $P(X \geq 5) = P(x = 5, 6, 7, 8, 9, \ldots)$

$$= \frac{1}{2^5} + \frac{1}{2^6} + \frac{1}{2^7} + \cdots$$

$a = \frac{1}{2^5} = \frac{1}{32}$

$r = \frac{1}{2}$

$\therefore P(x \geq 5) = \frac{\frac{1}{32}}{1-\frac{1}{2}} = \frac{2}{32} = \frac{1}{16}$

3) A random variable X has the following probability fn for various x.

x	0	1	2	3	4	5	6	7
p(x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

i) Find constant k.

ii) Evaluate $P(x \leq 6)$, $P(x \geq 6)$, $P(3 < x \leq 6)$.

iii)

4) A random expt of throwing a die twice is performed. A random variable X on this sample space is defined to be the sum of 2 nos. turning up. Find the discrete prob. distribution for the random variable X.

and compute the corresponding mean and std. deviation.

3) Soln:
since X is a discrete random variable, we must have

i) $p(x) \geq 0$, which is true only if $k \geq 0$.

ii) $\sum p(x) = 1$

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$10k^2 + 9k = 1$$

$$k(10k + 9) = 1$$

$k = 1$ and $10k + 9 = 1$

$$k = \frac{-8}{10} = \frac{-4}{5} < 0$$

$$10k^2 + 9k - 1 = 0$$

$$10k^2 + 10k - k - 1 = 0$$

$$10k(k+1) - 1(k+1) = 0$$

$k = -1$ and $10k - 1 = 0$

$$k = \frac{1}{10}$$

$$P(X < 6) = P(X = 0, 1, \ldots 5)$$

$$= 0 + k + 2k + 2k + 3k + k^2$$

$$= k^2 + 8k$$

$$= \frac{1}{100} + \frac{8}{10} = \frac{81}{100}$$

$$P(X \geq 6) = P(X = 6, 7) = 1 - P(X < 6) = 1 - \frac{81}{100} = \frac{19}{100}$$

$$P(3 < x \leq 6) = P(X = 4, 5, 6) = 3k + k^2 + 2k^2 = 3k + 3k^2 = 3k(1 + k)$$

$$= 3\left(\frac{1}{10}\right)\left(1 + \frac{1}{10}\right) = \frac{3}{10}\left(\frac{11}{10}\right) = \frac{33}{100}$$

4) Soln: $S = \{(x, y) : x, y = 1, 2, 3, 4, 5, 6\}$

i.e, $S = \{(1,1), (1,2), (1,3) \ldots (1,6), (2,1) \ldots (6,6)\}$

Total outcomes possible $n(S) = 36$

we have $X = \{2, 3, 4, \ldots 11, 12\}$ (sum)

$X = 2$ $(1,1)$ $P(X=2) = \frac{1}{36}$

$X = 3$ $(1,2), (2,1)$ $P(X=3) = \frac{2}{36}$

$X = 4$ $(1,3), (3,1), (2,2)$ $P(X=4) = \frac{3}{36}$

$X = 5$ $P(X=5) = \frac{4}{36}$

$X = 6$ $P(X=6) = \frac{5}{36}$

$X = 7$ $P(X=7) = \frac{6}{36}$

$8 = \frac{5}{36}$ $9 = \frac{4}{36}$ $10 = \frac{3}{36}$ $11 = \frac{2}{36}$ $12 = \frac{1}{36}$

$X = 12$ $(6,6)$ $P(X=6) = \frac{1}{36}$

$$\text{Mean}(\mu) = \sum x_i p(x_i)$$

$$= \frac{140}{36} + \frac{112}{36} = \frac{252}{36} = 7$$

$$\text{Variance} = 2\left(\frac{105}{36}\right) = 5.833 \rightarrow (x_i - \mu)^2 p_{x_i}$$

$$\text{std. dev} = 2.4152$$

1) If a random variable X take the values 1, 2, 3, 4 such that $2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4)$, find the probability distribution function of X.

2) If X is a discrete random variable taking values 0, 1, 2, 3, etc with $P(X) = \frac{1}{2}\left(\frac{2}{3}\right)^x$, find P(x being an odd no) by first establishing that P(x) is a probability function.

3) X is a discrete random variable having P(x) defined as follows

$$P(X) = \begin{cases} \frac{x}{15} & \text{if } 1 \le x \le 5 \\ 0 & : x > 5 \end{cases}$$ s.t. P(x) is a prob. function. Also find

i) $P(X = 1 \text{ or } 2)$ ii) $P\left(\frac{1}{2} < x < \frac{5}{2} \mid x > 1\right)$.

Soln:

1) Let $\{X, P(X)\}$ be the discrete probability distribution

X	1	2	3	4
P(X)	P_1	P_2	P_3	P_4

We have to find P_1, P_2, P_3, P_4 such that their sum is 1.

$$P_1 + P_2 + P_3 + P_4 = 1$$

Given:- $2P_1 = 3P_2 = P_3 = 5P_4$

$2P_1 = 3P_2 \Rightarrow P_2 = \frac{2}{3}P_1$

$2P_1 = 5P_4 \Rightarrow P_4 = \frac{2}{5}P_1$

$$P_1 + \frac{2}{3}P_1 + 2P_1 + \frac{2}{5}P_1 = 1$$

$$\Rightarrow 3P_1 + \frac{16}{15}P_1 = 1 \Rightarrow \frac{45P_1 + 16P_1}{15} = 1 \Rightarrow \frac{61}{15}P_1 = 1$$

$$\therefore \boxed{P_1 = \frac{15}{61}}$$

$$P_2 = \frac{2}{3}P_1 = \frac{2}{3} \times \frac{15}{61} = \frac{10}{61} \qquad P_3 = 2P_1 = 2 \times \frac{15}{61} = \frac{30}{61}$$

$$P_4 = \frac{2}{5}P_1 = \frac{2}{5} \times \frac{15}{61} = \frac{6}{61}$$

2) $P(x) = \frac{1}{2}\left(\frac{2}{3}\right)^x$ for $x = 1, 2, 3, \ldots$

clearly $P(x) \ge 0$

and consider $\sum_1^\infty P(x) = \frac{1}{2}\left(\frac{2}{3}\right)^1 + \frac{1}{2}\left(\frac{2}{3}\right)^2 + \frac{1}{2}\left(\frac{2}{3}\right)^3 + \ldots$

$$= \frac{1}{2}\left(\frac{2}{3}\right)\left[1 + \frac{2}{3} + \left(\frac{2}{3}\right)^2 + \ldots\right]$$

G.P. $a = 1$, $r = \frac{2}{3} < 1$

$= \frac{1}{2}\left(\frac{2}{3}\right) \cdot \frac{1}{\left(1-\frac{2}{3}\right)}$ $\quad \frac{a}{1-r}$

$= \frac{1}{3} \cdot \frac{3}{1}$

$\therefore \sum P(x) = 1$

$\therefore P(x)$ is probability function

$P(x) = \frac{1}{2}\left(\frac{2}{3}\right)^x$ for $x = 1, 2, 3, \ldots$

$P(x \text{ being odd number}) = \frac{1}{2}\left(\frac{2}{3}\right)^1 + \frac{1}{2}\left(\frac{2}{3}\right)^3 + \frac{1}{2}\left(\frac{2}{3}\right)^5 + \frac{1}{2}\left(\frac{2}{5}\right)^7 + \ldots$

$= \frac{1}{2}\left(\frac{2}{3}\right)\left[1 + \frac{4}{9} + \left(\frac{4}{9}\right)^2 + \ldots\right]$ G.P. $a = 1$; $r = \frac{4}{9} < 1$

$= \frac{1}{2}\left(\frac{2}{3}\right) \cdot \frac{1}{1-\frac{4}{9}}$

$= \frac{1}{3} \cdot \frac{9}{5}$

$= \frac{3}{5} = 0.6$

3) i) $P(X) \geq 0$

ii) Sum, $\sum_1^\infty P(x) = \sum_1^5 P(x) + \sum_5^\infty P(x)$

$= \frac{1}{15} + \frac{2}{15} + \frac{3}{15} + \frac{4}{15} + \frac{5}{15} + 0 + 0 + \ldots$

$= \frac{15}{15} + 0$

$= 1$

$\therefore P(x)$ is a probability function.

i) $P(X=1 \text{ or } 2) = P(X=1) + P(X=2)$

$= \frac{1}{15} + \frac{2}{15} = \frac{3}{15} = \frac{1}{5}$

ii) $P\left(\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right) = \frac{P\left(\frac{1}{2} < X < \frac{5}{2} \cap X > 1\right)}{P(X>1)} = \frac{P\left(\frac{1}{2} < X < \frac{5}{2}\right) \cdot P(X>1)}{P(X>1)}$

$=$

Bernoulli

Binomial distribution / Bernoulli theorem

A random expt with only 2 possible outcomes categorized as success and failure is called a Bernoulli trial, where the prob. of success 'p' is same for each trial.

Bernoulli's theorem :

The prob. of 'x' successes in 'n' trials is ${}^nC_x p^x q^{n-x}$ where,

p → prob. of success
q → prob of failure.

Proof: Since p is the prob. of success, the prob. of x successes is

$p.p.p.p. \ldots$ x times $= p^x$

Also x successes imply $(n-x)$ failures and since q is the prob of failure, the prob of $(n-x)$ failures is $q.q.q \ldots (n-x)$ times $= q^{n-x}$

By the multiplication rule, the prob. of simultaneous happening is

$p^x q^{n-x}$.

But x successes in n trials can occur in ${}^nC_x$ ways and all these cases are favourable to the event. Hence by the addition rule, the prob of x successes out of n trails is given by

$p^x q^{n-x} + p^x q^{n-x} + \ldots$ ${}^nC_x$ times $= {}^nC_x p^x q^{n-x}$

The prob. distribution $\{x, p(x)\}$ is given by

x	0	1	2	3 . . .	n-1	n
p(x)	q^n	${}^nC_1 p^1 q^{n-1}$	${}^nC_2 p^2 q^{n-2}$	. . .	${}^nC_{n-1} p^{n-1} q$	p^n

${}^nC_0 p^0 q^{n-0} + {}^nC_1 p^1 q^{n-1} + {}^nC_2 p^2 q^{n-2} + \ldots {}^nC_{n-1} p^{n-1} q^{n-(n-1)} + {}^nC_n p^n q^{n-n}$

$q^n + npq^{n-1} + {}^nC_2 p^2 q^{n-2} + \ldots + {}^nC_{n-1} p^{n-1} q + p^n$

$= (p+q)^n$

$= 1^n$

$= 1$

It may be observed that value of p^x for different values $x = 0, 1, 2, 3, \ldots n$ are the successive terms in the binomial expansn of $(q+p)^n$ and

accordingly this distribution is called the binomial distribution.

- The mean(μ) value of binomial distribution is np.
- The std. deviation of the , $\sigma = \sqrt{npq}$

1) The prob. that a person aged 60 years will live upto 70 years is 0.65. what is the prob. that out of 10 persons aged 60 atleast 7 of them will live upto 70?

$p = 0.65$ $\quad n = 10$ $\quad X$ – no. of persons aged 60 yrs living upto 70 years. $\quad {}^nC_x\, p^x q^{n-x}$

$q = 0.35$

$P(\text{atleast } 7) = P(X=7) + P(X=8) + P(X=9) + P(X=10)$

$= {}^{10}C_7 (0.65)^7 (0.35)^3 + {}^{10}C_8 (0.65)^8 (0.35)^2 + {}^{10}C_9 (0.65)^9 (0.35)^1 + {}^{10}C_{10} (0.65)^{10} (0.35)^0$

$= \frac{10 \times 9 \times 8}{3 \times 2} (0.049)(0.042) + \frac{10 \times 9}{2} (0.031)(0.1225) + 10 \times (0.02)(0.35) + 1 (0.013)$

$= 0.2469 + 0.17 + 0.07 + 0.013$

$= 0.499$

$P(X \geq 7) = 0.5138$ exact

2) An airline knows that 5% of the people making reservations on a certain flight will not turn up. Consequently, their policy is to sell 52 tickets for a flight that can only hold 50 people. What is the prob. that there will be a seat for every passenger who turns up?

Soln:- Let X be the no. of passengers who will not turn up.

$n = 52 \quad p = 0.05 \quad q = 1 - 0.05 = 0.95$

A seat is assured for every passenger who turns up if the no. of passengers who fail to turn up is more than or equal to 2.

$P(x) = {}^nC_x\, p^x q^{n-x}$

$= {}^{52}C_x (0.05)^x (0.95)^{52-x}$

$$P(x \geq 2) = \sum_{x=2}^{52} P(x)$$

$$P(x \geq 2) = 1 - \{ P(x=0) + P(x=1) \}$$

$$= 1 - \{ {}^{52}C_0 (0.05)^0 (0.95)^{52} + {}^{52}C_1 (0.05)(0.95)^{51} \}$$

$$= 1 - \{ 1 (0.95)^{52} + \frac{52!}{51!\,1!} (0.05)(0.95)^{51} \}$$

$$= 1 - 0.2594$$

$$= 0.7405$$

3) The no. of telephone lines busy at an instant of time is a binomial variate with prob ~~0.01~~ that a line is busy. If 10 lines are chosen at random, what is the prob. that

i) no line is busy

ii) all lines are busy

iii) atleast one line is busy

iv) almost 2 lines are busy

Soln:- Let x be the no. of telephone lines busy (which follows binomial distribution).

Wkt $P(x) = {}^nC_x\, p^x q^{n-x}$

$n=10 \quad p=0.1, \; q=1-0.1 = 0.9$

The prob that out of 10 telephone lines -

i) no line is busy

$$P(x=0) = {}^{10}C_0 (0.1)^0 (0.9)^{10}$$

$$= 0.34867$$

ii) All lines are busy

$$P(x=10) = {}^{10}C_{10} (0.1)^{10} (0.9)^0$$

$$= 1 \times 10^{-10}$$

iii) atleast one line is busy

$$P(x=1 \text{ to } x=10) = 1 - P(x<1)$$

$$= 1 - 0.3487$$

$$= 0.6513$$

iii) Atmost 2 lines are busy

$$P(x \le 2) = P(x=0) + P(x=1) + P(x=2)$$

$$= 0.3487 + {}^{10}C_1 (0.1)^1 (0.9)^{10-1} + {}^{10}C_2 (0.1)^2 (0.9)^8$$

$$= 0.3487 + 0.387 + 0.1937$$

$$= \underline{0.9294}$$

4) In a quiz contest of answering yes or no. What is the prob. of guessing atleast 6 answers correctly out of 10 questions asked? Also find the prob of the same if there are 4 options for a correct answer.

bihomial

e chosen

Soln:-

5) Find the bihomial prob distribution which has mean 2 and variance $\frac{4}{3}$.

Soln:- For bihomial distribution, we have

Mean, $\mu = np = 2$ — ①

variance, $(\sigma)^2 = npq = \frac{4}{3}$ — ②

②/① implies $\Rightarrow \frac{np}{npq} = \frac{2}{4/3} = \frac{1}{q} = \frac{3}{2}$

$\therefore \boxed{q = \frac{2}{3}}$

$1 - p = q \Rightarrow \underline{p = \frac{1}{3}}$

① $\Rightarrow np = 2 \Rightarrow n = 2 \times \frac{3}{1} = \underline{6}$

The prob distribution $p(x) = {}^nC_x \, p^x q^{n-x}$

$$= {}^6C_x \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{6-x}$$

x	0	1	2	3	4	5	6
$P(x)$	$(2/3)^6$	${}^6C_1\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^5$	${}^6C_2\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^4$	${}^6C_3\left(\frac{1}{3}\right)^3\left(\frac{2}{3}\right)^3$	${}^6C_4\left(\frac{1}{3}\right)^4\left(\frac{2}{3}\right)^2$	${}^6C_5\left(\frac{1}{3}\right)^5\left(\frac{2}{3}\right)^1$	${}^6C_6\left(\frac{1}{3}\right)^6$

6) In 800 families with 5 children each. How many families would be expected to have

i) 3 boys ii) 2 girls iii) either 2 or 3 boys

iv) Atmost 2 girls, by assuming prob. of girls & boys to be equal.

$(p = \frac{1}{2}, \; q = \frac{1}{2})$

Poisson Distribution

It is ~~divided~~ regarded as the limiting form of binomial distribution when 'n' is very large ($n \to \infty$) and the prob. of success 'p' is very small ($p \to 0$), so that np tends to a fixed finite constant 'm'.

The prob. of 'x' successes out of n trails is given by

$$P(x) = \frac{e^{-m} m^x}{\lfloor x}$$

$\lfloor x = x!$

The prob. distribution for $x = 0, 1, 2, \ldots$ etc is as follows.

x	0	1	2	3	...
$P(x)$	e^{-m}	$e^{-m}m$	$\frac{e^{-m}m^2}{\lfloor 2}$	$\frac{e^{-m}m^3}{\lfloor 3}$	...

We have $P(x) \geq 0$ &

$$\sum_{x=0}^{\infty} P(x) = e^{-m} + e^{-m}m + \frac{e^{-m}m^2}{\lfloor 2} + \frac{e^{-m}m^3}{\lfloor 3} + \ldots$$

$$= e^{-m}\left[1 + m + \frac{m^2}{\lfloor 2} + \frac{m^3}{\lfloor 3} + \ldots\right]$$

$$= e^{-m}\left[e^m\right]$$

$$= e^0$$

$$= 1$$

Hence $P(x)$ is a probability function of the discrete random variable X.

For a poisson distribution, the mean value $\mu = m$.

The std. deviation, $\sigma = \sqrt{m}$

Eg:- 1) A shop has 4 diesel generator sets which hires everyday. The demand for a generator set on an average is a poisson variate ($P(x)$) with value $\frac{5}{2}$. Obtain the prob. that on a particular day

i) there was no demand

ii) Demand had to be refused

Soln:- Let X be

Poisson dis

we have for a

Given: $\mu = m$

$P(x) =$

(i) Prob. tha

(ii) Prob that

If the

demand for

Thus $P(x >$

$= 1 - \{0.$

$= 1 - \{0.$

$= 1 - 0.$

$= 0.109$

3) The probab

equal. Calc

Given: $P(x =$

We $P(x$

$\frac{e^{-m} m^3}{3!}$

$\frac{1}{3!} =$

$m =$

Soln: Let X be the no. of demands for generator sets which follows Poisson distribution.

we have for a Poisson distribution, $P(x) = \frac{e^{-m} m^x}{\lfloor x}$ and the mean, $\mu = m$.

Given: $(\mu = m/2)$ $m = \frac{5}{2} = 2.5$

$$P(x) = \frac{e^{-2.5} (2.5)^x}{\lfloor x}$$

(i) Prob. that there was no demand is $P(x=0) = \frac{e^{-2.5}(2.5)^0}{0!}$

$$= 0.082$$

(ii) Prob that the demand had to refused

If the ——" ——, there should have been a demand for more than 4 generators.

Thus $P(x>4) = 1 - P(x \le 4)$

$$= 1 - \{ P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4) \}$$

$$= 1 - \left\{ 0.082 + \frac{e^{-2.5}(2.5)}{1!} + \frac{e^{-2.5}(2.5)^2}{2!} + \frac{e^{-2.5}(2.5)^3}{3!} + \frac{e^{-2.5}(2.5)^4}{4!} \right\}$$

$$= 1 - \{ 0.082 + 0.205 + 0.256 + 0.2137 + 0.1336 \}$$

$$= 1 - 0.8903$$

$$= 0.1097 \quad 0.1088$$

2) The probabilities of a Poisson variate taking the values 3 and 4 are equal. Calculate the prob. of the variate taking the values 0 and 1

Given: $P(x=3) = P(x=4)$

W.K.T $P(x) = \frac{e^{-m} m^x}{\lfloor x}$

$$\frac{e^{-m} m^3}{3!} = \frac{e^{-m} m^4}{4!}$$

$$\frac{1}{3!} = \frac{m}{4 \times 3!}$$

$$m = 4$$

$$P(x=0) = \frac{e^{-m} m^x}{x!}$$

$$= \frac{e^{-4} m^0}{0!}$$

$$= 0.0183$$

$$P(x=1) = \frac{e^{-4} m^1}{1!} = e^{-4}(4)$$

$$= 0.0732$$

3) Probability that a news reader commits no mistake in reading the news is $\frac{1}{e^3}$. Find the prob. that on a particular news broadcast he commits

i) only 2 mistakes ii) more than 3 mistakes 0.3528

iii) almost 3 mistakes. 0.6472

Soln:- Let x be the no. of mistakes committed by newsreader.

Prob. that there are no mistakes $= P(x=0) = \frac{1}{e^3}$

We $P(x) = \frac{e^{-m} m^x}{x!}$

$P(x=0) = \frac{e^{-m} \cdot m^0}{0!}$

$\frac{1}{e^3} = e^{-m} \Rightarrow e^{-3} = e^{-m} \Rightarrow \underline{m=3}$

i) only 2 mistakes

$x=2 \quad P(x=2) = \frac{e^{-3}(3)^2}{2!} = \underline{0.224}$

ii) more than 3 mistakes

$P(x>3) = 1 - P(x \leq 3)$

$= 1 - \{P(x=0) + P(x=1) + P(x=2) + P(x=3)\}$

$= 1 - \{e^{-3} + \frac{e^{-3}(3)^1}{1!} + \frac{e^{-3}(3)^2}{2!} + \frac{e^{-3}(3)^3}{3!}\}$

$= 1 - \{0.0497 + 0.1493 + 0.224 + 0.224\}$

$= 1 - 0.647$

$= \underline{0.353}$

iii) $P(x \leq 3) = \underline{0.647}$

4) The no. of accidents in a year to taxi drivers in a city follows a Poisson distribution with mean 3. Out of 1000 taxi drivers, find approximately the no. of drivers with

i) no accident in a year ii) more than 3 accidents in a year

Soln:- Let x be the no. of accidents to a taxi driver in a year which follows Poisson distribution.

Given: $\mu = 3 = m$

i) $P(x=0) = \frac{e^{-m} m^0}{0!} = \frac{e^{-3}}{1} = 0.0497$

(ii) more than 3 accidents in a year

$$P(x > 3) = 1 - \{P(x=0) + P(x=1) + P(x=2) + P(x=3)\}$$

$$= 1 - \{0.0497 + 0.1493 + 0.224 + 0.224\}$$

$$= 1 - 0.647$$

$$= 0.353$$